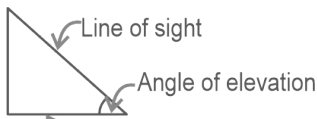


Height and Distance

In this chapter following concepts / terminologies will be used.

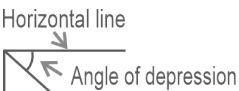
Angle Of Elevation



Horizontal Line

The angle made by the line of sight with the horizontal line is called the angle of elevation.

Angle Of Depression



The angle made by the line of sight with the horizontal line is called the angle of depression.

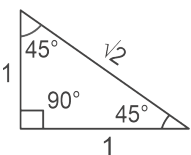
Note: When we come near object, the angle of elevation increases and when we go away from the object, the angle of elevation decreases.

Note: Angle of elevation and angle of depression are always equal.

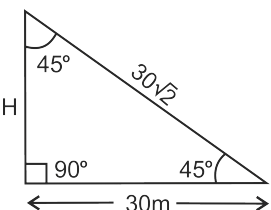
In this chapter trigonometric ratio and concepts will be used which we have discussed in detail in the trigonometry chapter.

Some important relation

Case :- (1)



Example :- The angle of elevation of the top of a tower from a point on the ground, 45° which is 30 m from the foot of the tower. Find the height of the tower?



Let the height of the tower be 'H'

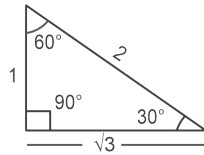
$$\tan 45^\circ = \frac{H}{30}$$

$$1 = \frac{H}{30} \Rightarrow H = 30 \text{ m}$$

Short Trick :- $1 \rightarrow 30 \text{ m}$

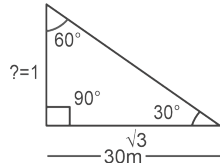
$$1 \rightarrow ? \Rightarrow ? = 30 \text{ m}$$

Case :- (2)



Example :- The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower is 30° . Find the height of the tower?

Solution :-



Let, the height of the tower be 'H'

$$\tan 30^\circ = \frac{H}{30}$$

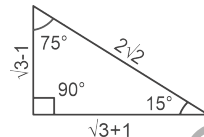
$$\frac{1}{\sqrt{3}} = \frac{H}{30} \Rightarrow H = \frac{30}{\sqrt{3}} \text{ m} = 10\sqrt{3} \text{ m}$$

Short Trick :-

$$\sqrt{3} \rightarrow 30 \text{ m}$$

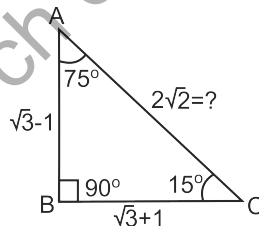
$$1 \rightarrow ? \Rightarrow ? = \frac{30}{\sqrt{3}} \text{ m} = 10\sqrt{3} \text{ m}$$

Case :- (3)



Example :- The angle of elevation of a ladder leaning against a wall is 15° and the foot of the ladder is 4 m away from the wall. The length of the ladder is :

Solution :-



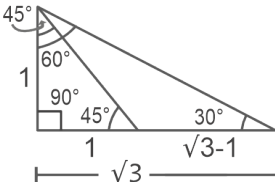
Let AB be the wall and AC be the ladder.

Then, $\angle ACB = 15^\circ$ and $BC = 4 \text{ m}$.

$$\sqrt{3} + 1 \rightarrow 4 \text{ m}$$

$$2\sqrt{2} \rightarrow ? \Rightarrow ? = \frac{8\sqrt{2}}{\sqrt{3} + 1}$$

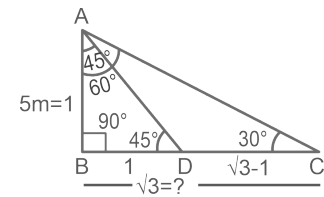
Case :- (4)



Example :- A man on the top of a vertical observation tower observes a car moving at a uniform speed coming directly towards it and the angle of depression to change from 30° to 45° . If the height of the tower is 5 m then find the value of the total covered distance by

the car?

Solution :-



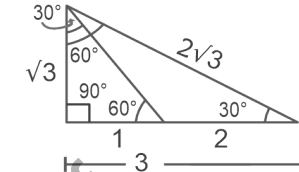
Let $AB = \text{Height of tower} = 1$ And total covered distance by the car $= BC = \sqrt{3}$

So, ATQ

$$1 = 5 \text{ m}$$

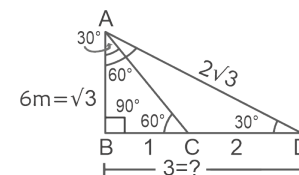
$$\sqrt{3} = ? \Rightarrow 5\sqrt{3}$$

Case :- (5)



Example :- A man on the top of a vertical observation tower observes a car moving at a uniform speed coming directly towards it and the angle of depression to change from 30° to 60° . If the height of the tower is 6 m then find the value of the total covered distance by the car?

Solution :-



Let $AB = \text{Height of tower} = \sqrt{3}$

And total covered distance by the car $= BD = 3$

So, A/Q

$$\sqrt{3} = 6 \text{ m}$$

$$3 = ? \Rightarrow 6\sqrt{3} \text{ m}$$

Variety Questions

Q.1. A man looks at the reflection of the top of the lamp-post on the mirror that is 6.6 m away from the foot of the lamppost. The man's height is 1.25 m and he is standing 2 m away from the mirror. Assuming that the mirror is placed on the ground, facing the sky and the man, and that the mirror and the lamp-post are in a same line, find the height of the lamp-post (in metres).

SSC CHSL 04/07/2024 (1st Shift)

(a) 3.97 (b) 4.28 (c) 4.45 (d) 4.13

Q.2. The shadow of a tower standing on level ground is found to be 40 m longer when the Sun's altitude is 30° than when it was 45° . Find the height (in meters) of the tower.

SSC CHSL Tier II 10/01/2024

- (a) $10(\sqrt{3} - 1)$ (b) $20(\sqrt{3} - 1)$
 (c) $20(\sqrt{3} + 1)$ (d) $10(\sqrt{3} + 1)$

Q.3. Two pillars of equal height stand on either side of a roadway which is 150 m wide. At a point in the road between pillars, the elevations of the pillars are x° and y° so that $\tan x^\circ = \frac{2}{5}$, $\tan y^\circ = \frac{3}{5}$,

then the height of each pillar is :

SSC CGL Tier II (26/10/2023)

- (a) 39 m (b) 36 m (c) 42 m (d) 33 m

Q.4. Subhas, a 3.15 m tall tree, and an 11.25 m tall building are positioned such that their feet on the ground are collinear and the tree is located between Subhas and the building. The tree is located at a distance of 7.5 m from Subhas and at a distance of 45 m from the building. Further, the eyes of Subhas, the top of the tree, and the top of the building fall in one line. Find the height (in m) from the ground at which Subhas' eyes are situated.

SSC CHSL 09/08/2023 (4th Shift)

- (a) 1.75 (b) 1.8 (c) 1.6 (d) 1.5

Q.5. From the top of an upright pole 17.75 m high, the angle of elevation of the top of an upright tower was 60° . If the tower was 57.75 m tall, how far away (in m) from the foot of the pole was the foot of the tower?

SSC CGL Tier II (03/03/2023)

- (a) $40\sqrt{3}$ (b) $\frac{151\sqrt{3}}{6}$ (c) $\frac{77}{4}\sqrt{3}$ (d) $\frac{40\sqrt{3}}{3}$

Q.6. A pole 23 m long reaches a window which is $3\sqrt{5}$ m above the ground on one side of a street. Keeping its foot at the same point, the pole is turned to the other side of the street to reach a window $4\sqrt{15}$ m high. What is the width (in m) of the street?

SSC CGL 11/04/2021 (Afternoon)

- (a) 17 (b) 35 (c) 39 (d) 22

Q.7. A person was standing on a road near a mall. He was 1215 m away from the mall and able to see the top of the mall from the road in such a way that the top of a tree, which is in between him and the mall, was exactly in line of sight with the top of the mall. The tree height is 20m and it is 60m away from him. How tall (in m) is the mall?

SSC CPO 25/11/2020 (Morning)

- (a) 405 (b) 250 (c) 300 (d) 375

Q.8. A ladder leaning against a wall makes an θ with the horizontal ground such that $\cos \theta = \frac{5}{13}$. If the height of the

top of the ladder from the wall is 18m, then what is the distance (in cm) of the foot of the ladder from the wall?

SSC CPO 23/11/2020 (Morning)

- (a) 19.5 (b) 13 (c) 7.5 (d) 18

Q.9. A pole stands vertically on a road, which goes in the north-south direction. P, Q are two points towards the north of the pole, such that PQ = b, and the angles of elevation of the top of the pole at P, Q are α , β respectively. Then the height of the pole is:

SSC CPO 09/12/2019 (Morning)

- (a) $\frac{b}{\tan \beta + \tan \alpha}$ (b) $\frac{b}{\cot \beta - \cot \alpha}$
 (c) $\frac{b}{\tan \beta - \tan \alpha}$ (d) $\frac{b \tan \alpha}{\tan \beta}$

Q.10. A clock tower stands at the crossing of two roads which point in the north-south and the east-west directions. P, Q, R and S are points on the roads due north, east, south and west respectively, where the angles of elevation of the top of the tower are respectively, α , β , γ and δ . Then $(\frac{PQ}{RS})^2$ is equal to:

SSC CPO 09/12/2019 (Morning)

- (a) $\frac{\tan^2 \alpha + \tan^2 \delta}{\tan^2 \gamma + \tan^2 \beta}$ (b) $\frac{\tan^2 \alpha + \tan^2 \beta}{\tan^2 \gamma + \tan^2 \delta}$
 (c) $\frac{\cot^2 \alpha + \cot^2 \beta}{\cot^2 \gamma + \cot^2 \delta}$ (d) $\frac{\cot^2 \alpha + \cot^2 \delta}{\cot^2 \gamma + \cot^2 \beta}$

Q.11. From a point exactly midway between the foot of two towers P and Q, the angles of elevation of their tops are 30° and 60° respectively. The ratio of the height of P to that of Q is:

SSC CGL Tier II (12/09/2019)

- (a) 1 : 3 (b) 1 : 2 (c) 1 : $2\sqrt{3}$ (d) 2 : $3\sqrt{3}$

Q.12. The angle of elevation of a flying drone from a point on the ground is 60° . After flying for 5 seconds the angle of elevation drops to 30° . If the drone is flying horizontally at a constant height of $1000\sqrt{3}$ m, the distance (in m) travelled by the drone is:

SSC CPO 16/03/2019 (Afternoon)

- (a) 2000 (b) 1000 (c) 3000 (d) 4000

Q.13. At a certain time of a day a tree 5.4 m height casts a shadow of 9 m. If a pole casts a shadow of 13.5 m at the same time, the height of the pole is:

SSC CPO 16/03/2019 (Morning)

- (a) 8.1 m (b) 9.9 m (c) 7.2 m (d) 6.3 m

Q.14. What is the angle of elevation of the sun from the top of a pillar, when its height is equal to the length of its shadow?

SSC CPO 15/03/2019 (Evening)

- (a) 30° (b) 90° (c) 45° (d) 60°

Q.15. The top of a broken tree touches the ground at a 60° angle, 45 meters away from the root of the tree. What will be the total height (in m) of the tree? (Use $\sqrt{3} = 1.73$ and $\sqrt{2} = 1.41$)

SSC CPO 14/03/2019 (Evening)

- (a) 153.45 (b) 141.3 (c) 167.85 (d) 137.24

Q.16. From the top of the 120 m high lighthouse, the angle of depression of two ships on opposite sides of the base of the lighthouse is 30° and 60° . What is the distance between the ships? (rounded off)

SSC CPO 14/03/2019 (Morning)

- (a) 327 m (b) 127 m (c) 277 m (d) 177 m

Q.17. From the top of 75 m high tower, the angle of depression of two points P and Q on opposite side of the base of the tower on level ground is θ and ϕ , such that $\tan \theta = \frac{3}{4}$ and $\tan \phi = \frac{5}{8}$. What is the distance (in m) between the points P and Q?

SSC CPO 13/03/2019 (Morning)

- (a) 190 (b) 200 (c) 180 (d) 220

Q.18. From the top of a 12 m high building, the angle of elevation of the top of the tower is 60° and the angle of depression of the foot of the tower is θ , such that $\tan \theta = \frac{3}{4}$. What is the height

of the tower? ($\sqrt{3} = 1.73$)

SSC CPO 12/03/2019 (Evening)

- (a) 41.41 (b) 36.22 (c) 39.68 (d) 37.95

Q.19. A ladder leaning against a wall makes an angle α with the horizontal ground such that $\tan \alpha = \frac{3}{4}$. If the foot of the ladder is 5 m away from the wall, what is the length (in m) of the ladder?

SSC CPO 12/03/2019 (Morning)

- (a) 5.25 (b) 3.75 (c) 6.25 (d) 4.5

Practice Questions

SSC CGL 2024 Tier - 1

Q.20. A pole 10 m long rests slantly against a vertical wall AB making an angle 30° with the horizontal (ground). Find how far the foot of the pole is from the wall (in metres).

SSC CGL 10/09/2024 (1st Shift)

- (a) 15 (b) $\frac{5}{\sqrt{3}}$ (c) 10 (d) $5\sqrt{3}$

SSC CHSL 2024 Tier - 1

Q.21. A mirror is placed on the ground facing upwards. A man sees the top of a tower in the mirror which is at a distance of 105 m from the mirror. The man is 0.5 m away from the mirror, and his height is 1.5 m. Find the height of the tower (in metres).

SSC CHSL 10/07/2024 (1st Shift)

(a) 280 (b) 115 (c) 315 (d) 210

Q.22. A vertical pole of length 6 m casts a 4 m long shadow on the ground. At the same time, a tower casts a 28 m long shadow. Find the height of the tower.

SSC CHSL 11/07/2024 (2nd Shift)

(a) 48 m (b) 32 m (c) 36 m (d) 42 m

SSC CHSL 2023 Tier - 2

Q.23. A pole stands vertically on the ground with the help of a 12-metre steel wire tied to its top and affixed on the ground. If the steel wire makes an angle of 30° with the horizontal ground, then the height of the pole is equal to:

SSC CHSL Tier II 02/11/2023

(a) 8 meters (b) $6\sqrt{3}$ meters
(c) $8\sqrt{3}$ meters (d) 6 meters

Q.24. A man observed the top of an electric pole at angles of elevation of 60° and 30° , respectively, from two different observation points P and Q. If the distance between P and Q is 10 m, find the height of the pole (in m).

SSC CHSL Tier II 10/01/2024

(a) $5\sqrt{3}$ (b) 5 (c) 15 (d) $15\sqrt{3}$

SSC CGL 2023 Tier - 2

Q.25. From the top of a tower, the angle of depression of the top of a 10 m high building is 60° . If the distance between the tower and the building is $50\sqrt{3}$ m, find the height of the tower.

SSC CGL Tier II (26/10/2023)

(a) 150 m (b) 100 m (c) 160 m (d) 140 m

SSC CHSL 2023 Tier - 1

Q.26. A vertical pillar 42 cm long casts a 35 cm long shadow. At the same time, a tower casts a shadow 25 m long. Find the height of the tower.

SSC CHSL 07/08/2023 (4th Shift)

(a) 28 m (b) 35 m (c) 32 m (d) 30 m

Q.27. Suhas, a 3.15 m tall tree, and a building are positioned such that their feet on the ground are collinear and the tree is located between Suhas and the building. The tree is located at a distance

of 7.5 m from Suhas and at a distance of 45 m from the building. Further, the eyes of Suhas, the top of the tree, and the top of the building fall in one line, and the eyes of Suhas are at a height of 1.8 m from the ground. Find the height (in m) of the building.

SSC CHSL 14/08/2023 (2nd Shift)

(a) 11.25 (b) 11.50 (c) 10.25 (d) 10.75

Q.28. Suhas, a tree, and an 11.25 m tall building are positioned such that their feet on the ground are collinear and the tree is located between Suhas and the building. The tree is located at a distance of 7.5 m from Suhas and at a distance of 45 m from the building. Further, the eyes of Suhas, the top of the tree, and the top of the building fall in one line, and the eyes of Suhas are at a height of 1.8 m from the ground. Find the height (in m) of the tree.

SSC CHSL 17/08/2023 (2nd Shift)

(a) 3.00 (b) 3.30 (c) 3.15 (d) 3.45

SSC CGL 2023 Tier - 1

Q.29. A person standing at a distance looks at a building having a height of 1000 metres. The angle between the top of the building and the ground is 30° . At what approximate distance (in metres) is the person standing away from the building.

SSC CGL 17/07/2023 (4th shift)

(a) 1000 (b) 936 (c) 1732 (d) 1542

Q.30. A vertical pole of 28m height casts a 19.2m long shadow. At the same time, find the length of the shadow cast by another pole of 52.5m height.

SSC CGL 20/07/2023 (4th shift)

(a) 36 m (b) 35 m (c) 40 m (d) 30 m

SSC CGL 2022 Tier - 2

Q.31. Two ships are on the opposite side of a lighthouse such that all three of them are collinear. The angles of depression of the two ships from the top of the lighthouse are 30° and 60° . If the ships are $230\sqrt{3}$ m apart, then find the height of the light house (in m).

SSC CGL Tier II (02/03/2023)

(a) 175.4 (b) 165.2 (c) 172.5 (d) 180.5

Q.32. From the top of an upright pole $24\sqrt{3}$ feet high, the angle of elevation of the top of an upright tower was 60° . If the foot of the pole was 60 feet away from the foot of the tower, what tall (in feet) was the tower?

SSC CGL Tier II (06/03/2023)

(a) $84\sqrt{3}$ (b) $36\sqrt{3}$ (c) $44\sqrt{3}$ (d) $60\sqrt{3}$

Q.33. The angle of elevation of the top of a tower from the top of a building whose height is 680 m is 45° and the angle of elevation of the top of the same tower from the foot of the same building is 60° . What is the height (in m) of the tower?

SSC CGL Tier II (07/03/2023)

(a) $340(3 + \sqrt{3})$ (b) $310(3 - \sqrt{3})$
(c) $310(3 + \sqrt{3})$ (d) $340(3 - \sqrt{3})$

SSC CGL 2022 Tier - 1

Q.34. An airplane goes 14 km due east and then 48 km due north. How far is it from its initial position?

SSC CGL 13/12/2022 (3rd Shift)

(a) 14 km (b) 48 km (c) 50 km (d) 28 km

SSC CGL 2021 Tier - 2

Q.35. A person 1.8 metre tall is $30\sqrt{3}$ metres away from a tower. If the angle of elevation from his eye to the top of the tower is 30° , then what is the height (in m) of the tower?

SSC CGL Tier II (08/08/2022)

(a) 32.5 (b) 37.8 (c) 30.5 (d) 31.8

SSC CHSL 2021 Tier - 1

Q.36. The foot of a ladder 25 m long is 7 m from the base of the building. If the top of the ladder slips by 4 m, then by how much distance will the foot of the ladder slide?

SSC CHSL 24/05/2022 (Morning)

(a) 9 m (b) 7 m (c) 6 m (d) 8 m

SSC CGL 2021 Tier - 1

Q.37. The angle of elevation of the top of an unfinished tower at a point distant 78 m from its base is 30° . How much higher must the tower be raised (in m) so that the angle of elevation of the top of the finished tower at the same point will be 60° ?

SSC CGL 11/04/2022 (Morning)

(a) $78\sqrt{3}$ (b) 80 (c) $52\sqrt{3}$ (d) $26\sqrt{3}$

Q.38. A vertical pole and a vertical tower are on the same level ground in such a way that, from the top of pole, the angle of elevation of the top of the tower is 60° and the angle of depression of the bottom of the tower is 30° . If the height of the pole is 24 m, then find the height of the tower (in m).

SSC CGL 12/04/2022 (Morning)

(a) $24\sqrt{3}(\sqrt{3} + 1)$ (b) 72
(c) 96 (d) $24(\sqrt{3} + 1)$

Q.39. The angle of elevation of the top of

a tall building from the points M and N at the distances of 72m and 128m, respectively, from the base of the building and in the same straight line with it, are complementary. The height of the building (in m) is?

SSC CGL 12/04/2022 (Afternoon)

(a) 84 (b) 96 (c) 80 (d) 90

Q.40. Exactly midway between the foot of two towers P and Q, the angle of elevation of their tops are 45° and 60° , respectively. The ratio of the heights of P and Q is:

SSC CGL 18/04/2022 (Morning)

(a) $1:\sqrt{3}$ (b) $3:1$ (c) $1:3$ (d) $\sqrt{3}:1$

Q.41. A poster is on top of a building. A person is standing on the ground at a distance of 50m from the building. The angles of elevation to the top of the poster and bottom of the poster are 45° and 30° , respectively. What is 200% of the height (in m) of the poster?

SSC CGL 18/04/2022 (Afternoon)

(a) $\frac{25}{3}(3-\sqrt{3})$ (b) $\frac{75}{3}(3-\sqrt{3})$
(c) $\frac{50}{3}(3-\sqrt{3})$ (d) $\frac{100}{3}(3-\sqrt{3})$

SSC CGL 2020 Tier - 2

Q.42. Two pillars A and B of the same height are on opposite sides of a road which is 40 m wide. The angles of elevation of the tops of the pillars A and B are 30° and 45° , respectively, at a point on the road between the pillars. What is the distance (in m) of the point from the foot of pillar A?

SSC CGL Tier II (03/02/2022)

(a) $40(\sqrt{3}-1)$ (b) $20(2-\sqrt{3})$
(c) $20(3-\sqrt{3})$ (d) $39\sqrt{3}$

SSC CHSL 2020 Tier - 1

Q.43. As observed from the top of a lighthouse, 42 m high above sea level, The angle of depression of a ship sailing directly towards it changes from 30° to 45° . The distance travelled by the ship during the period of observation is:

SSC CHSL 04/08/2021 (Evening)

(a) $42(1-\sqrt{3})$ (b) $42(\sqrt{3}+1)$
(c) $42(\sqrt{3}-1)$ (d) 42

SSC CPO 2020 Tier - 1

Q.44. A person was standing on a road near a mall. He was 1425 m away from the mall and able to see the top of the mall from the road in such a way that the top of the tree, which is in between him and the mall, was exactly in line of sight with the top of the mall. The tree height

is 10m and it is 30m away from him. How tall (in m) is the mall?

SSC CPO 23/11/2020 (Morning)

(a) 475 (b) 300 (c) 425 (d) 525

Q.45. The length of the shadow of a vertical pole on the ground is 18m. If the angle of elevation of the sun at that time is θ , such that $\cos\theta = \frac{12}{13}$, then what is the height (in m) of the pole?

SSC CPO 23/11/2020 (Evening)

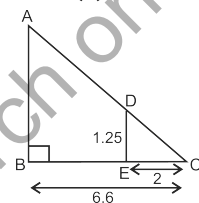
(a) 18 (b) 9 (c) 7.5 (d) 12

Answer Key :-

1.(d)	2.(c)	3.(b)	4.(b)
5.(d)	6.(c)	7.(a)	8.(c)
9.(b)	10.(c)	11.(a)	12.(a)
13.(a)	14.(c)	15.(c)	16.(c)
17.(d)	18.(c)	19.(c)	20.(d)
21.(c)	22.(d)	23.(d)	24.(a)
25.(c)	26.(d)	27.(a)	28.(c)
29.(c)	30.(a)	31.(c)	32.(a)
33.(a)	34.(c)	35.(d)	36.(d)
37.(c)	38.(c)	39.(b)	40.(a)
41.(d)	42.(c)	43.(c)	44.(a)
45.(c)			

Solutions :-

Sol.1.(d)

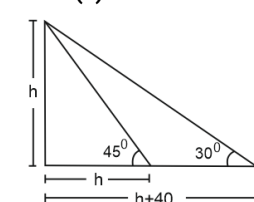


From similarity concept :- $\frac{EC}{ED} = \frac{BC}{AB}$

$$\frac{2}{1.25} = \frac{6.6}{AB}$$

(height of the lamp post) $AB = 3.3 \times 1.25 = 4.125$ m (4.13 approx)

Sol.2.(c)



$$\tan 30^\circ = \frac{h}{h+40}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{h+40}$$

$$h+40 = \sqrt{3}h$$

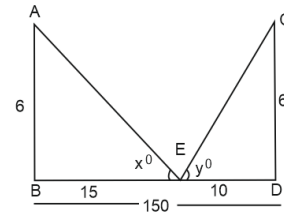
$$\therefore \text{height of tower (h)} = \frac{40}{\sqrt{3}-1}$$

$$= 20(\sqrt{3}+1) \text{ meter}$$

Sol.3.(b) Let AB and CD be the two pillars of equal height

$$\tan x^\circ = \frac{AB}{BE} = \frac{2}{5} \text{ or } \frac{6}{15}$$

$$\tan y^\circ = \frac{CD}{DE} = \frac{3}{5} \text{ or } \frac{6}{10}$$

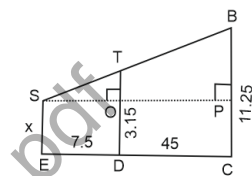


ATQ,

$$15 + 10 = 25 \text{ unit} \text{ ----- } 150 \text{ m}$$

$$\text{Then, 6 unit} \text{ ----- } \frac{150}{25} \times 6 = 36 \text{ m}$$

Sol.4.(b)



$$EC = SP = 7.5 + 45 = 52.5 \text{ metre}$$

$$BP = 11.25 - x \text{ metre}$$

$$\Delta SOT \sim \Delta SPB$$

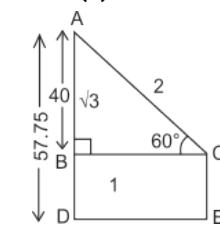
$$\frac{SO}{SP} = \frac{TO}{BP} \Rightarrow \frac{7.5}{52.5} = \frac{TO}{BP} \Rightarrow \frac{TO}{BP} = \frac{1}{7}$$

$$6 \text{ units} = (11.25 - 3.15) = 8.1 \text{ metre}$$

$$TO (1 \text{ unit}) = \frac{8.1}{6} = 1.35 \text{ metre}$$

$$SE = OD = (3.15 - 1.35) = 1.8 \text{ meter}$$

Sol.5.(d)



Angle $\rightarrow 30^\circ : 60^\circ : 90^\circ$

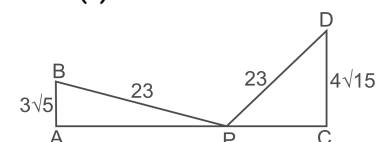
Side $\rightarrow 1 : \sqrt{3} : 2$

In ΔABC ,

$$\sqrt{3} \text{ units} = 40 \text{ m} \Rightarrow 1 \text{ unit} = \frac{40\sqrt{3}}{3} \text{ m}$$

So, The foot of the tower is $\frac{40\sqrt{3}}{3}$ m away from the foot of the pole.

Sol.6.(c)

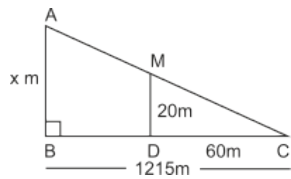


Let P be the fixed point of the pole and B and D are two windows.,

$$\text{In } \Delta ABP, AP = \sqrt{(23)^2 - (3\sqrt{5})^2} = \sqrt{529 - 45} = \sqrt{484} = 22 \text{ m.}$$

$$\text{In } \Delta DPC, PC = \sqrt{(23)^2 - (4\sqrt{15})^2} = \sqrt{529 - 240} = \sqrt{289} = 17 \text{ m.}$$

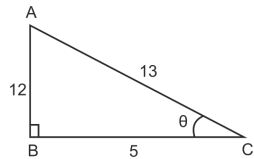
So, width of street = $22 + 17 = 39$ m

Sol.7.(a)

we have to find out the length of AB

$\triangle ABC \sim \triangle MDC$ (by AAA rule)

$$\frac{x}{20} = \frac{1215}{60} \Rightarrow x = \frac{1215 \times 20}{60} = 405 \text{ m}$$

Sol.8.(c)

It is given that

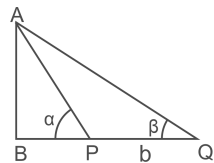
$$\cos \theta = \frac{\text{base}}{\text{hypotenuse}} = \frac{BC}{AC} = \frac{5}{13}$$

Using pythagoras theorem,

we have $AB = 12$ unit

ATQ , 12 unit = 18 m

$$\text{Then, } 5 \text{ unit} = \frac{18}{12} \times 5 = 7.5 \text{ m}$$

Sol.9.(b) Let the pole length be AB

In $\triangle ABP$, we have ;

$$\tan \alpha = \frac{AB}{BP} \Rightarrow BP = AB \cot \alpha$$

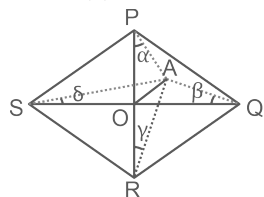
In $\triangle ABQ$, we have ;

$$\tan \beta = \frac{AB}{BQ} \Rightarrow BQ = AB \cot \beta$$

Now, $b = BQ - BP$

$$b = AB \cot \beta - AB \cot \alpha = AB(\cot \beta - \cot \alpha)$$

$$AB = \frac{b}{\cot \beta - \cot \alpha}$$

Sol.10.(c)

Let OA be a straight tower.

P, Q, R and S are points on the roads due north, east, south and west respectively.

In $\triangle POQ$ and $\triangle ORS$

$$PQ^2 = PO^2 + OQ^2$$

$$RS^2 = OR^2 + OS^2$$

$$\frac{(PQ)^2}{(RS)^2} = \frac{PO^2 + OQ^2}{OR^2 + OS^2} \text{ ----- (1)}$$

Here, $\angle AOQ = \angle AOP = \angle AOR = \angle AOS = 90^\circ$ (as AO is a straight tower)

Now,

the angle of elevation are α, β, γ and δ .

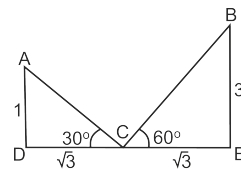
$$\text{In } \triangle AOP, \tan \alpha = \frac{OA}{OP} \Rightarrow OP = OA \cot \alpha$$

$$\text{In } \triangle AOS, \tan \delta = \frac{OA}{OS} \Rightarrow OS = OA \cot \delta$$

Similarly, $OQ = AO \cot \beta$ and $OR = AO \cot \gamma$

Putting the value of PO, OQ, OR, OS in eqn (1), we have

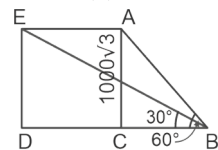
$$\frac{(PQ)^2}{(RS)^2} = \frac{(AO \cot \alpha)^2 + (AO \cot \beta)^2}{(AO \cot \gamma)^2 + (AO \cot \delta)^2} = \frac{\cot^2 \alpha + \cot^2 \beta}{\cot^2 \gamma + \cot^2 \delta}$$

Sol.11.(a)

Let the height of tower P and Q be AD and BE respectively and C be the midpoint of DE.

From the given figure, we have ;

Required ratio = $AD : BE = 1 : 3$

Sol.12.(a)

In $\triangle ACB$, we have :

$$\tan 60^\circ = \frac{1000\sqrt{3}}{BC} \Rightarrow \sqrt{3} = \frac{1000\sqrt{3}}{BC}$$

$$BC = 1000 \text{ m}$$

In $\triangle EDB$, we have :

$$\tan 30^\circ = \frac{1000\sqrt{3}}{BD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{1000\sqrt{3}}{BD}$$

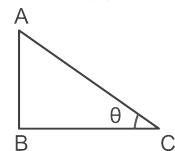
$$BD = 3000 \text{ m}$$

$$\text{Distance covered by drone} = BD - BC = 3000 - 1000 = 2000 \text{ m}$$

Sol.13.(a) Let the height of the pole be h.

$$\text{ATQ, } \frac{h}{13.5} = \frac{5.4}{9}$$

$$9h = 72.9 \Rightarrow h = 8.1 \text{ m}$$

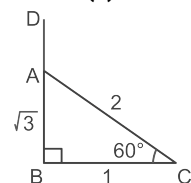
Sol.14.(c)

Let AB and BC be the length of pole and the length of shadow respectively

$AB = BC$ (given)

$$\tan \theta = \frac{AB}{BC} = 1 = \tan 45^\circ \Rightarrow \theta = 45^\circ$$

Short trick : If the height of an object and length of its shadow is equal. Then, the angle is 45°

Sol.15.(c)

Let BD be the total height of the tree.

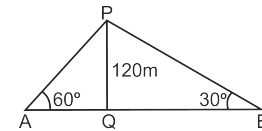
$AD = AC$ (given)

Here, 1 unit = 45 m

$$\text{Then, } \sqrt{3} \text{ unit} = 45\sqrt{3} \text{ m}$$

$$\text{and } 2 \text{ unit} = 45 \times 2 = 90 \text{ m}$$

$$\text{So, the total height of tree} = 45\sqrt{3} + 90 = 77.85 + 90 = 167.85 \text{ m}$$

Sol.16.(c)

$$\text{In } \triangle APQ, \text{ we have ; } \tan 60^\circ = \frac{PQ}{AQ}$$

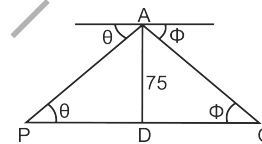
$$\sqrt{3} = \frac{120}{AQ} \Rightarrow AQ = \frac{120}{\sqrt{3}} = 40\sqrt{3}$$

$$\text{In } \triangle BPQ, \text{ we have ; } \tan 30^\circ = \frac{PQ}{QB}$$

$$\frac{1}{\sqrt{3}} = \frac{120}{QB} \Rightarrow QB = 120\sqrt{3}$$

Therefore,

$$AB = 40\sqrt{3} + 120\sqrt{3} = 160\sqrt{3} \approx 277 \text{ m}$$

Sol.17.(d)

Let AD be the height of the tower.

$$\tan \theta = \frac{AD}{PD} = \frac{3}{4}, \tan \phi = \frac{AD}{DQ} = \frac{5}{8}$$

Here, 3 unit = 75 m

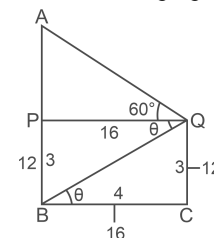
$$\text{Then, } 4 \text{ unit} = \frac{75}{3} \times 4 = 100 \text{ m}$$

Again, 5 unit = 75 m

$$\text{Then, } 8 \text{ unit} = \frac{75}{5} \times 8 = 120 \text{ m}$$

$$\text{So, } PQ = 100 + 120 = 220 \text{ m}$$

Sol.18.(c) As per the directions given, we have following figure :

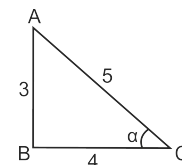


$$\text{In } \triangle AQP, \tan 60^\circ = \frac{AP}{16} \Rightarrow \sqrt{3} = \frac{AP}{16}$$

$$AP = 16\sqrt{3} = 16 \times 1.73 = 27.68$$

$$\text{So, height of tower} = 27.68 + 12 = 39.68 \text{ m}$$

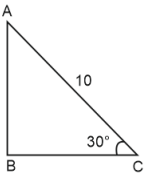
Sol.19.(c) Let AB and AC be the height of wall and ladder respectively



Here, 4 unit = 5

$$\text{Then, } 5 \text{ unit} = \frac{5}{4} \times 5 = \frac{25}{4} = 6.25 \text{ m}$$

Sol.20.(d) Let AB and AC be the length of wall and pole respectively.



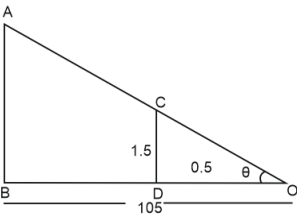
In $\triangle ABC$, $\cos 30^\circ = \frac{BC}{10}$

$$\frac{\sqrt{3}}{2} = \frac{BC}{10}$$

$$BC = \frac{10\sqrt{3}}{2} = 5\sqrt{3} \text{ m}$$

So, the foot of the pole is $5\sqrt{3}$ m away from the wall.

Sol.21.(c) Let AB be the height of tower and CD be the height of person and Mirror is placed at O.



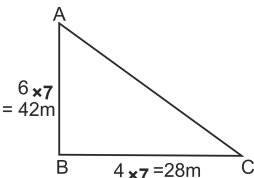
$\triangle OCD \sim \triangle OAB$ (By AAA rule)

$$\frac{CD}{AB} = \frac{OD}{OB} \Rightarrow \frac{1.5}{AB} = \frac{0.5}{105}$$

$$AB = \frac{1.5 \times 105}{0.5} = 315 \text{ m}$$

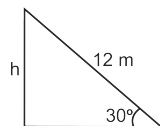
Sol.22.(d)

Let AB be the length of the pole and BC be the shadow on the ground.



So, the height of tower = 42 m

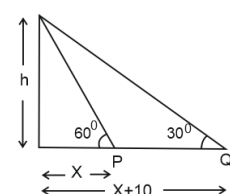
Sol.23.(d)



$$\sin 30^\circ = \frac{h}{12}$$

$$\frac{1}{2} = \frac{h}{12}, h = 6 \text{ m}$$

Sol.24.(a)



$$\tan 60^\circ = \frac{h}{x}$$

$$h = \sqrt{3}x \text{(i)}$$

$$\tan 30^\circ = \frac{h}{x+10}$$

$$\sqrt{3}h = x + 10$$

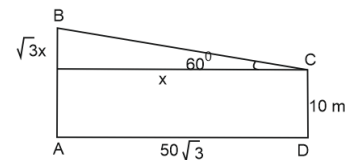
$$h = \frac{x+10}{\sqrt{3}} \text{(ii)}$$

From equations (i) and (ii) we get

$$\sqrt{3}x = \frac{x+10}{\sqrt{3}}$$

$$\therefore x = 5 \Rightarrow h = 5\sqrt{3} \text{ m}$$

Sol.25.(c) Let AB and CD be the height of tower and building respectively.



Height of tower = $(\sqrt{3}x + 10)$ m

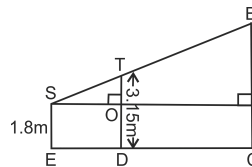
$$\text{ATQ, } x = 50\sqrt{3}$$

$$\text{Then, height of tower} = (\sqrt{3} \times 50\sqrt{3} + 10) = 150 + 10 = 160 \text{ m}$$

Sol.26.(d)

$$\frac{42}{35} = \frac{x}{25} \Rightarrow x = \frac{42 \times 25}{35} = 30 \text{ m}$$

Sol.27.(a)



$$TO = TD - OD = 3.15 - 1.8 = 1.35 \text{ m}$$

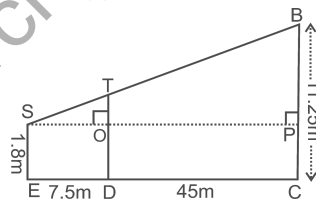
$\triangle SOT \cong \triangle SPB$

$$\frac{SO}{SP} = \frac{TO}{BP} \Rightarrow \frac{7.5}{45 + 7.5} = \frac{1.35}{BP}$$

$$BP = \frac{1.35 \times 52.5}{7.5} = 9.45 \text{ m}$$

$$\text{Height of the building} = (BP + PC) = 9.45 + 1.8 = 11.25 \text{ m}$$

Sol.28.(c)



$$PB = BC - PC = 11.25 - 1.8 = 9.45 \text{ m}$$

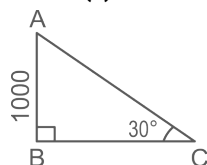
$\triangle SOT \cong \triangle SPB$

$$\frac{SO}{SP} = \frac{TO}{BP} \Rightarrow \frac{7.5}{7.5 + 45} = \frac{TO}{9.45}$$

$$TO = \frac{7.5 \times 9.45}{52.5} = 1.35 \text{ m}$$

$$\text{Length of tree (TD)} = TO + OD = 1.35 + 1.8 = 3.15 \text{ m}$$

Sol.29.(c)



$$\tan 30^\circ = \frac{1000}{BC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{1000}{BC}$$

$$BC = 1000\sqrt{3} \text{ m} = 1000 \times 1.732 = 1732 \text{ m}$$

Alternate method :

Ratio of length of side corresponding to

$$30^\circ, 60^\circ \text{ and } 90^\circ = 1 : \sqrt{3} : 2$$

$$1 \text{ unit} = 1000 \text{ m}$$

(distance between building and man)

$$\sqrt{3} \text{ units} = 1000 \times 1.732 = 1732 \text{ m}$$

Sol.30.(a)

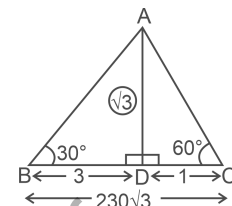
Shadow of pole $\propto$ height of pole

A pole of 28 m height casts a shadow of 19.2 m long .

52.5 m long pole casts a shadow of

$$= \frac{19.2}{28} \times 52.5 = 36 \text{ metre}$$

Sol.31.(c) Let B and C be two ships and AD be the lighthouse



In triangle ADC

$$\text{Angle} \rightarrow 30^\circ : 60^\circ : 90^\circ$$

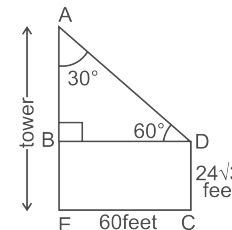
$$\text{Side} \rightarrow 1 : \sqrt{3} : 2$$

$$4 \text{ units} = 230\sqrt{3} \text{ m}$$

Height of the light house $(\sqrt{3} \text{ units})$

$$= \frac{230\sqrt{3}}{4} \times \sqrt{3} = \frac{690}{4} = 172.5 \text{ meters.}$$

Sol.32.(a)



$$\text{Angle} \rightarrow 30^\circ : 60^\circ : 90^\circ$$

$$\text{Side} \rightarrow 1 : \sqrt{3} : 2$$

$$BD = EC = 60 \text{ feet and } DC$$

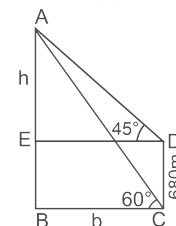
$$= BE = 24\sqrt{3} \text{ feet} \Rightarrow AB = \sqrt{3}, BD = 1$$

According to the questions ,

$$1 \text{ unit} = 60 \text{ feet} \Rightarrow \sqrt{3} \text{ units} = 60\sqrt{3} \text{ feet}$$

$$\text{Height of tower (AE)} = 60\sqrt{3} + 24\sqrt{3} = 84\sqrt{3} \text{ feet}$$

Sol.33.(a)



Let AB be the length of the tower and CD be the length of the building

$$BE = CD = 680 \text{ m}$$

$$AE = h \text{ meter and } DE = b \text{ meter}$$

Then, In $\triangle AED$

$$\tan 45^\circ = \frac{h}{b} = 1 \text{ (so } b = h)$$

Then, In $\triangle ABC$

$$\tan 60^\circ = \frac{h + 680}{b} = \sqrt{3}$$

$$h + 680 = \sqrt{3} b (\because b = h)$$

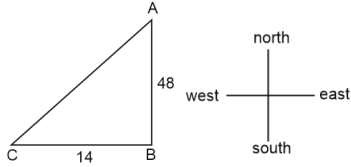
$$680 = h(\sqrt{3} - 1) \Rightarrow h = \frac{680}{(\sqrt{3}-1)} \times \frac{(\sqrt{3}+1)}{(\sqrt{3}-1)}$$

$$h = \frac{680}{2} \times (\sqrt{3} + 1) \Rightarrow h = 340(\sqrt{3} + 1)$$

$$\Rightarrow AB = h + BE \Rightarrow AB = 340(1 + \sqrt{3}) + 680$$

$$AB = 340(3 + \sqrt{3}) \text{ m}$$

Sol.34.(c) Applying the directions given in the question, we have following figure:

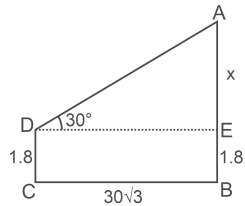


$$AC = \sqrt{48^2 + 14^2} = \sqrt{2304 + 196} = \sqrt{2500} = 50 \text{ km}$$

Sol.35.(d) Let the AB be a tower and $AE = x$, so $AB = (1.8 + x)$

$$DE = CB = 30\sqrt{3} \text{ m}$$

According to the question,

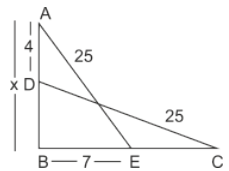


In $\triangle ADE$,

$$\tan 30^\circ = \frac{AE}{DE} \Rightarrow \frac{1}{\sqrt{3}} = \frac{x}{30\sqrt{3}} \Rightarrow x = 30 \text{ m}$$

$$\text{So, height of tower} = 1.8 + x = 1.8 + 30 = 31.8 \text{ m}$$

Sol.36.(d)



Length of the ladder $AE = DC = 25 \text{ m}$

Distance of foot from the base (BE) = 7 m

In triangle ABE,

$$AB^2 + BE^2 = AE^2 \Rightarrow x^2 + 7^2 = 25^2$$

$$x = \sqrt{625 - 49} = 24 \text{ m}$$

$$DB = 24 - 4 = 20 \text{ m}$$

Now in triangle DBC,

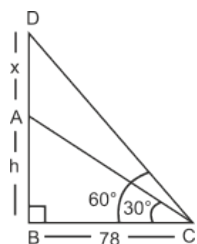
$$DB^2 + BC^2 = DC^2 \Rightarrow 20^2 + BC^2 = 25^2$$

$$BC^2 = 625 - 400 = 225 \Rightarrow BC = \sqrt{225} = 15 \text{ m}$$

$$EC = BC - BE = 15 - 7 = 8 \text{ m}$$

Hence, the foot of the ladder will slide by 8 m.

Sol.37.(c) Let AB be the height of tower and the AD be the height of the tower to be raised



$$\text{In } \triangle ABC, \tan 30^\circ = \frac{h}{78}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{78} \Rightarrow h = 26\sqrt{3}$$

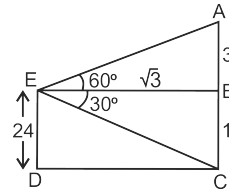
In $\triangle DBC$,

$$\tan 60^\circ = \frac{26\sqrt{3} + x}{78} \Rightarrow \sqrt{3} = \frac{26\sqrt{3} + x}{78}$$

$$78\sqrt{3} = 26\sqrt{3} + x \Rightarrow x = 52\sqrt{3} \text{ m}$$

$$\text{So, the required height of tower} = 52\sqrt{3} \text{ m}$$

Sol.38.(c) Let ED and AC be the height of pole and tower respectively
ATQ, we have following figure;

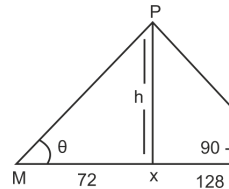


Here, 1 unit ----- 24 m

$$\text{Then, } (3 + 1) = 4 \text{ unit} \text{ ----- } 24 \times 4 = 96 \text{ m}$$

$$\text{So, the height of tower} = 96 \text{ m}$$

Sol.39.(b) Let the height of building be h

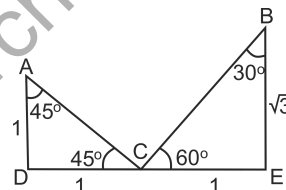


when angles are complementary

$h^2 = \text{Product of the distance of two points from the building}$

$$h = \sqrt{72 \times 128} = \sqrt{9216} = 96 \text{ m}$$

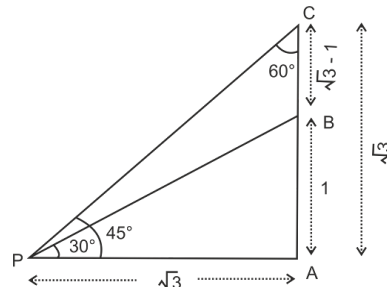
Sol.40.(a) Let the height of tower P and Q be AD and BE respectively and C be the midpoint of DE.



From the given figure, we have ;

$$\text{The ratio of the heights of P and Q} = 1 : \sqrt{3}$$

Sol.41.(d) Let the height of the building be AB and that of the poster be BC.
and "P" be the position of the standing person. we get following figure



According to the question,

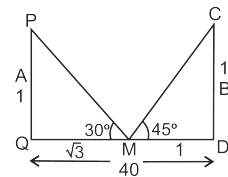
$$\sqrt{3} \text{ unit} = 50 \text{ m}$$

$$(\sqrt{3} - 1) \text{ unit} = \frac{50}{\sqrt{3}} \times (\sqrt{3} - 1) \text{ m}$$

$$\frac{50}{\sqrt{3}} \times (\sqrt{3} - 1) \times \frac{\sqrt{3}}{\sqrt{3}} \Rightarrow \frac{50}{3} (3 - \sqrt{3})$$

$$200\% \text{ of the poster} = \frac{100}{3} (3 - \sqrt{3}) \text{ m.}$$

Sol.42.(c)



Let the height of pillars A and B be PQ and CD respectively and M be the point on the road between the pillars.

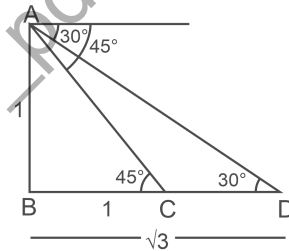
According to the question,

$$(\sqrt{3} + 1) \text{ unit} = 40 \text{ m}$$

$$\sqrt{3} \text{ unit} = \frac{40 \times \sqrt{3}}{(\sqrt{3} + 1)}$$

$$= \frac{40 \times \sqrt{3}(\sqrt{3} - 1)}{(\sqrt{3} + 1)(\sqrt{3} - 1)} = 20(3 - \sqrt{3}) \text{ m}$$

Sol.43.(c)



From the given figure, we have;

$$CD = BD - BC = (\sqrt{3} - 1) \text{ unit}$$

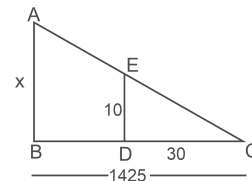
$$\text{Here, 1 unit} \text{ ----- } 42 \text{ m}$$

$$\text{Then, } (\sqrt{3} - 1) \text{ unit} \text{ ----- } 42(\sqrt{3} - 1) \text{ m}$$

So, the distance travelled by the ship during the period of observation

$$= 42(\sqrt{3} - 1) \text{ m}$$

Sol.44.(a)

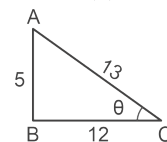


From the given figure, we can clearly see

$$\text{that } \frac{10}{x} = \frac{30}{1425} \text{ (As, } \triangle CED \sim \triangle CAB)$$

$$\Rightarrow x = \frac{1425 \times 10}{30} = 475 \text{ m}$$

Sol.45.(c)



It is given that

$$\cos \theta = \frac{\text{base}}{\text{hypotenuse}} = \frac{BC}{AC} = \frac{12}{13}$$

Using pythagoras theorem, we have $AB = 5 \text{ unit}$

$$\text{ATQ, } 12 \text{ unit} \rightarrow 18 \text{ m}$$

$$\text{Then, } 5 \text{ unit} \rightarrow \frac{18}{12} \times 5 = 7.5 \text{ m}$$